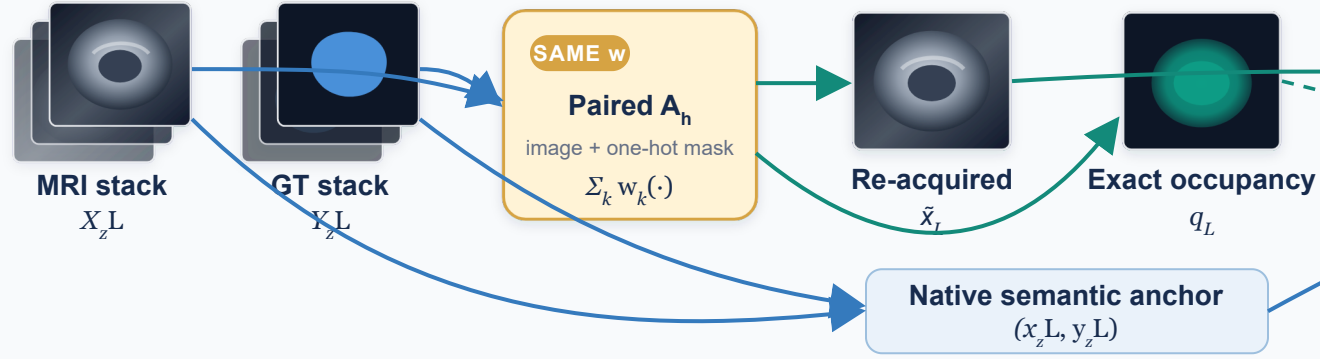
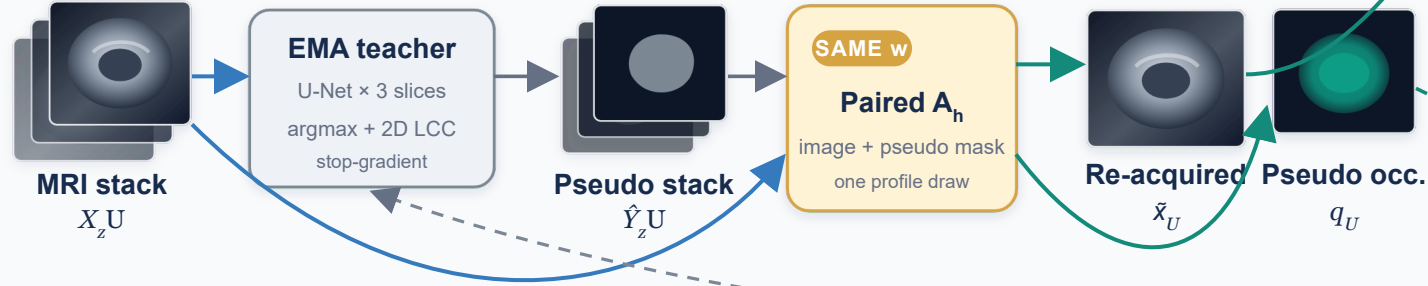


(a) Paired slice re-acquisition

LABELED

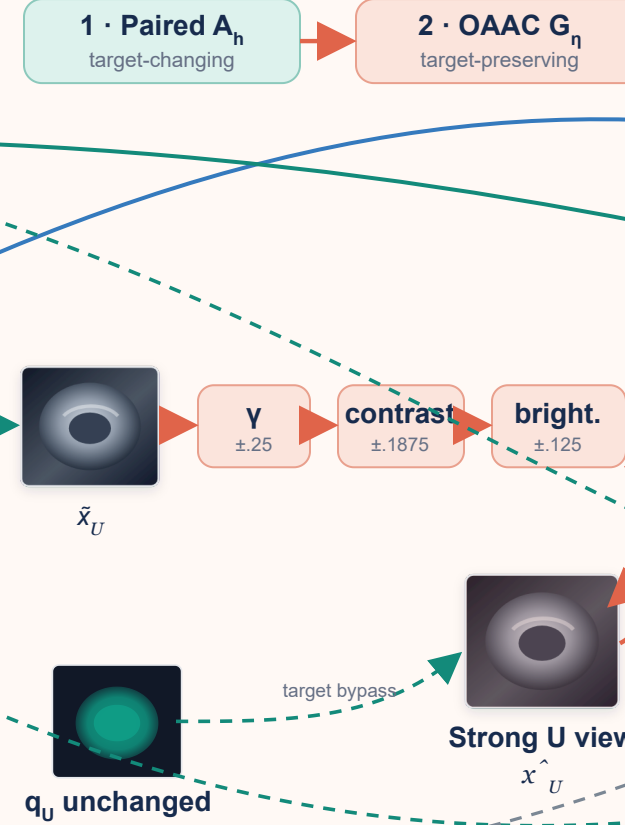


UNLABELED

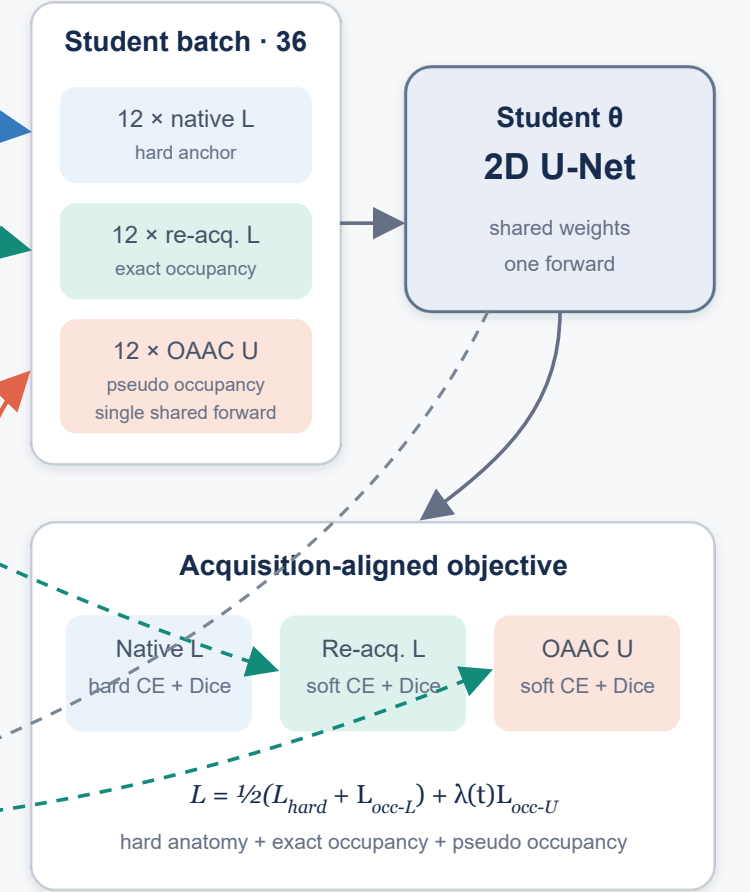


EMA update: $\xi \leftarrow \alpha \xi + (1-\alpha) \theta$

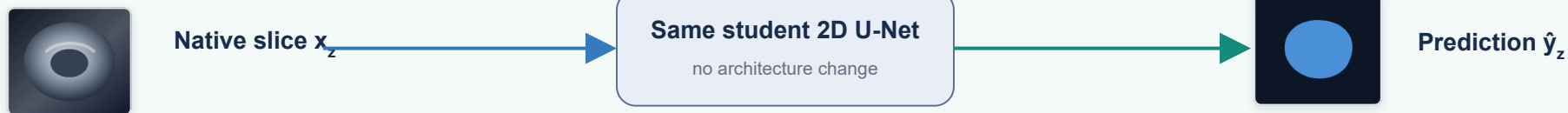
(b) Ordered appearance



(c) Shared learner



(d) Inference · unchanged single-slice path



Training-only modules are removed

no neighboring slices · no A_h · no OAAC
no extra parameters or inference latency